

Cricket Player Performance Tracking Application

Built on the Salesforce Platform

Salesforce Administrator with AI (Agentforce) Program

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1. Introduction

Cricket teams, academies and clubs generate a huge amount of data every match: runs, balls faced, wickets, catches and more. When this data is kept in notebooks or scattered Excel files, coaches find it hard to compare players, identify form and take selection decisions.

The **Cricket Player Performance Tracking Application** is a no-code / low-code solution built on the **Salesforce platform**. It stores players, teams, matches and match-wise performance in custom objects, calculates batting average, strike rate and economy automatically, and presents the results in reports and dashboards.

Why Salesforce?

- Declarative tools (Object Manager, Flow, Reports) let the app be built quickly without heavy coding.
- Built-in security, user management and mobile access through the Salesforce mobile app.
- Free **Developer Edition** org is available for learning and building projects.
- AI features such as Agentforce can be added to answer questions about player stats in natural language.

2. Salesforce Developer Account Creation

Before building the application, a free **Salesforce Developer Edition** org is needed. It is a full-featured environment meant for learning, development and testing. Estimated time: about 20 minutes (Complexity: Easy).

Step-by-Step Procedure

Step	Action
1	Open a browser and go to the Salesforce Developer Edition sign-up page (developer.salesforce.com/signup).
2	Fill the form: First Name, Last Name, Email, Role, Company, Country and Postal Code.
3	Enter a Username in email format (e.g. yourname@cricketdemo.com). It must be globally unique; it need not be a real mailbox.
4	Accept the Master Subscription Agreement and click Sign me up .
5	Open the verification email sent to the address you entered and click Verify Account .
6	Set a strong password and a security question, then log in.
7	Confirm you reach the Salesforce Home page in Lightning Experience .
8	Click the gear icon and open Setup to see Object Manager, Flows and Users.

Tips & Common Problems

- If the username is already taken, add a number or a different word to make it unique.
- Check the spam folder if the verification email does not arrive.
- Bookmark your org's My Domain login URL and keep the username safely; the email address is used only for verification and password reset.
- Do not use a production or client org for practice. Always build in the Developer org.
- The sign-up screens may change slightly over time, but the required details stay the same.

3. Problem Statement & Objectives

3.1 Problem Statement

Coaches and selectors need a quick and reliable way to review how each player is performing across matches. Manual score sheets are time-consuming, easily lost and do not support comparison, trend analysis or instant reports. A centralised, automated tracking system is required.

3.2 Objectives

- Create a central database for teams, players and matches in Salesforce.
- Record match-wise batting, bowling and fielding statistics for every player.
- Automatically calculate strike rate, batting average and economy rate.
- Show career totals on each player record using roll-up summary fields.
- Build reports and dashboards such as top run-scorers and leading wicket-takers.
- Restrict data entry and editing using profiles and permission sets.

3.3 Scope

The application covers player profiles, team management, match scheduling, performance entry and analytics. Live ball-by-ball scoring, fantasy features and ticketing are outside the current scope.

3.4 Users

User	Role in the application
Admin	Manages objects, users and security
Coach / Analyst	Enters match performance and views dashboards
Selector	Reads reports to shortlist players
Player (optional)	Views own statistics through a community or app page

4. Requirements & Salesforce Features Used

4.1 Functional Requirements

ID	Requirement	Priority
FR1	Add and edit teams with coach and home ground	High
FR2	Add and edit players with role, batting and bowling style	High
FR3	Create matches with date, venue, format and result	High
FR4	Record each player's performance per match	High
FR5	Auto-calculate strike rate, economy and batting average	High
FR6	Reports and dashboards for team and player analysis	Medium
FR7	Validation to prevent wrong data (e.g. negative runs)	Medium

4.2 Salesforce Features Used

Feature	Purpose in this project
Custom Objects & Fields	Store teams, players, matches and performances
Lookup / Master-Detail	Connect the objects and enable roll-up summaries
Formula Fields	Strike rate, economy rate, batting average
Roll-Up Summary Fields	Career runs, wickets and matches on Player
Validation Rules	Data quality checks
Lightning App Builder	Custom app, tabs and record pages
Reports & Dashboards	Visual analysis for coaches
Flow Builder	Automate updates and notifications
Profiles & Permission Sets	Access control
Agentforce (AI)	Natural-language questions about player stats

5. Data Model (Custom Objects)

Four custom objects form the core of the application. API names end with **__c** as per Salesforce convention.

5.1 Player__c

Field	Type	Description
Player Name	Text (Name)	Full name of the player
Role__c	Picklist	Batter, Bowler, All-rounder, Wicket-keeper
Batting_Style__c	Picklist	Right-hand / Left-hand
Bowling_Style__c	Picklist	Fast, Medium, Off-spin, Leg-spin, None
Jersey_Number__c	Number	Unique jersey number
Team__c	Lookup(Team)	Team the player belongs to
Total_Runs__c	Roll-Up (SUM)	Sum of Runs_Scored__c from performances
Total_Wickets__c	Roll-Up (SUM)	Sum of Wickets__c from performances

5.2 Team__c and Match__c

Object	Key Fields
Team__c	Name, Coach__c (Text), Home_Ground__c (Text), Country__c (Picklist)
Match__c	Match_Date__c (Date), Venue__c (Text), Format__c (T20 / ODI / Test), Team_A__c and Team_B__c (Lookup), Result__c (Picklist)

5.3 Match_Performance__c (Junction Object)

Field	Type	Description
Player__c	Master-Detail	Parent: Player
Match__c	Master-Detail	Parent: Match
Runs_Scored__c, Balls_Faced__c	Number	Batting figures
Fours__c, Sixes__c	Number	Boundaries
Overs_Bowled__c, Runs_Conceded__c, Wickets__c	Number	Bowling figures
Catches__c	Number	Fielding figures
Strike_Rate__c	Formula	$\text{Runs_Scored_c} / \text{Balls_Faced_c} * 100$
Economy_Rate__c	Formula	$\text{Runs_Conceded_c} / \text{Overs_Bowled_c}$

6. Schema Diagram & Relationships

The schema below shows how the objects are connected. The Match Performance object sits between Player and Match, allowing one player to have many performances and one match to have many player performances.

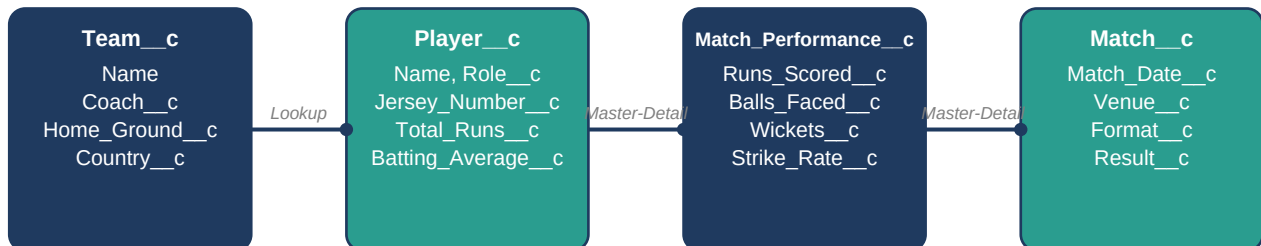


Figure 1: Data model (schema) of the Cricket Player Performance Tracking Application

Match_Performance__c is a junction object linking Player and Match (many-to-many).

Relationship Summary

Relationship	Type	Reason
Team to Player	Lookup	A player can exist without a team (free agent) and can move teams.
Player to Match Performance	Master-Detail	Performance cannot exist without a player; enables roll-up summary.
Match to Match Performance	Master-Detail	Performance belongs to one match; deleting the match removes its records.
Team to Match	Lookup (two fields)	Every match has Team A and Team B.

Formulas Used

Batting Average = $\text{Total_Runs_c} / \text{Times_Out_c}$

Strike Rate = $\text{IF}(\text{Balls_Faced_c} > 0, \text{Runs_Scored_c} / \text{Balls_Faced_c} * 100, 0)$

Economy = $\text{IF}(\text{Overs_Bowled_c} > 0, \text{Runs_Conceded_c} / \text{Overs_Bowled_c}, 0)$

7. Application Build

The application is assembled using point-and-click tools in Salesforce Setup.

Step	Where in Setup	What is done
1	Object Manager	Create the four custom objects with tabs enabled
2	Fields & Relationships	Add fields, picklist values, lookup and master-detail relationships
3	Page Layouts / Lightning Record Pages	Arrange fields; add related lists (e.g. Performances on Player)
4	App Manager	Create the Lightning app "Cricket Tracker" with tabs: Teams, Players, Matches, Performances, Reports, Dashboards
5	Validation Rules	Block invalid data
6	Data Import Wizard	Load sample teams and players from CSV
7	Reports	Create summary and joined reports
8	Dashboards	Add charts and tables from the reports

Sample Validation Rules

Rule	Error Condition	Message
Runs_Not_Negative	$\text{Runs_Scored_c} < 0$	Runs cannot be negative.
Fours_Sixes_Limit	$(\text{Fours_c} * 4 + \text{Sixes_c} * 6) > \text{Runs_Scored_c}$	Boundary runs exceed total runs.
Overs_Limit	$\text{Overs_Bowled_c} > 10$ for T20 matches	T20 bowler cannot bowl more than 4 overs.

Reports & Dashboard Components

- Top 10 Run Scorers (bar chart) from Player report sorted by Total Runs.
- Leading Wicket-Takers (table) grouped by player.
- Team-wise average strike rate (donut / column chart).
- Recent match performances with a date filter (last 30 days).

8. Automation, Security & AI (Agentforce)

8.1 Automation with Flow

- **Record-Triggered Flow:** when a Match Performance is created with 50+ runs or 3+ wickets, tag it as a Milestone and notify the coach.
- **Screen Flow:** a simple form to enter a full match scorecard for all players in one place.
- **Scheduled Flow:** weekly email with the top performers to selectors.

8.2 Security

User Type	Profile / Permission Set	Access
Admin	System Administrator	Full access
Coach / Analyst	Cricket Coach (custom)	Create/Edit performances; read all objects
Selector	Cricket Selector (custom)	Read-only, reports and dashboards

Organisation-Wide Defaults are set to **Private** for Player and Match_Performance so that only permitted users can view or edit sensitive statistics, and sharing rules open records to selectors.

8.3 AI with Agentforce

Agentforce allows an AI agent to be created inside Salesforce that can understand plain-language questions and act on the org's data. In this project an agent can be configured with topics and actions so that a coach can ask questions such as:

- "Who scored the most runs in the last five T20 matches?"
- "Show the bowling economy of our spinners this season."
- "Summarise Player X's form and suggest whether to include him in the playing XI."

Agent actions can be built using Flow or Apex that query the Player and Match Performance objects. Availability of Agentforce features depends on the org's edition and licences, so check Setup before enabling it.

9. Testing, Conclusion & References

9.1 Test Cases

ID	Scenario	Expected Result	Status
TC1	Create a player with Team selected	Player saved and visible on Team page	Pass
TC2	Enter performance: 60 runs, 40 balls	Strike Rate shows 150	Pass
TC3	Enter negative runs	Validation error displayed	Pass
TC4	Add performances for a player	Total Runs roll-up updates	Pass
TC5	Selector tries to edit performance	Access denied	Pass
TC6	Run Top Run Scorers report	Correct sorted list shown	Pass

9.2 Conclusion

The Cricket Player Performance Tracking Application shows how Salesforce can be used beyond CRM to solve a real sports-analytics problem. With a properly designed data model, formulas, automation and dashboards, coaches get instant, accurate insights into player form without manual calculation.

9.3 Future Enhancements

- Experience Cloud site so players and fans can view statistics.
- Integration with live-score APIs to load match data automatically.
- Einstein / Agentforce predictions for player form and injury risk.
- Fitness and training modules linked to each player.

9.4 References

- Salesforce Help and Trailhead: Data Modeling, Formulas and Validation Rules, Flow Builder.
- Salesforce Developer documentation: Object Reference and Metadata concepts.
- Salesforce Agentforce documentation on Salesforce Help.